

Sampling Distributions

Sampling Distribution



Simple Idea:

Distribution of a statistic

Compare to the distribution of a given variable



Formal Definition:

Sampling distribution is the distribution of values taken by a statistic across all possible samples of a given size, drawn from a particular population

All statistics have a sampling distribution

$\bar{x}$, mode, third quartile, upper extreme, etc

Terminology:

- $\mu_x \sigma_x$
 - Raw score distribution
- $\mu_{\bar{x}} \sigma_{\bar{x}} \mu_{\bar{y}}$
 - Parameters of the sampling distribution of that particular statistic

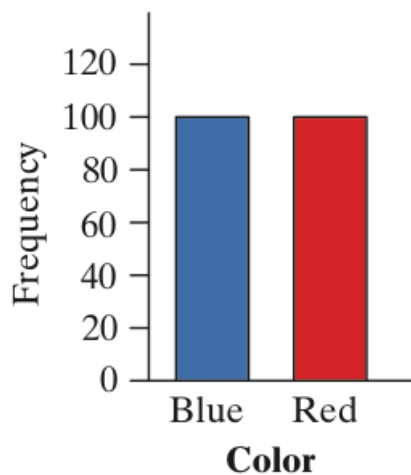


Population Distributions are different

Be careful: The population distribution describes the individuals that make up the population.

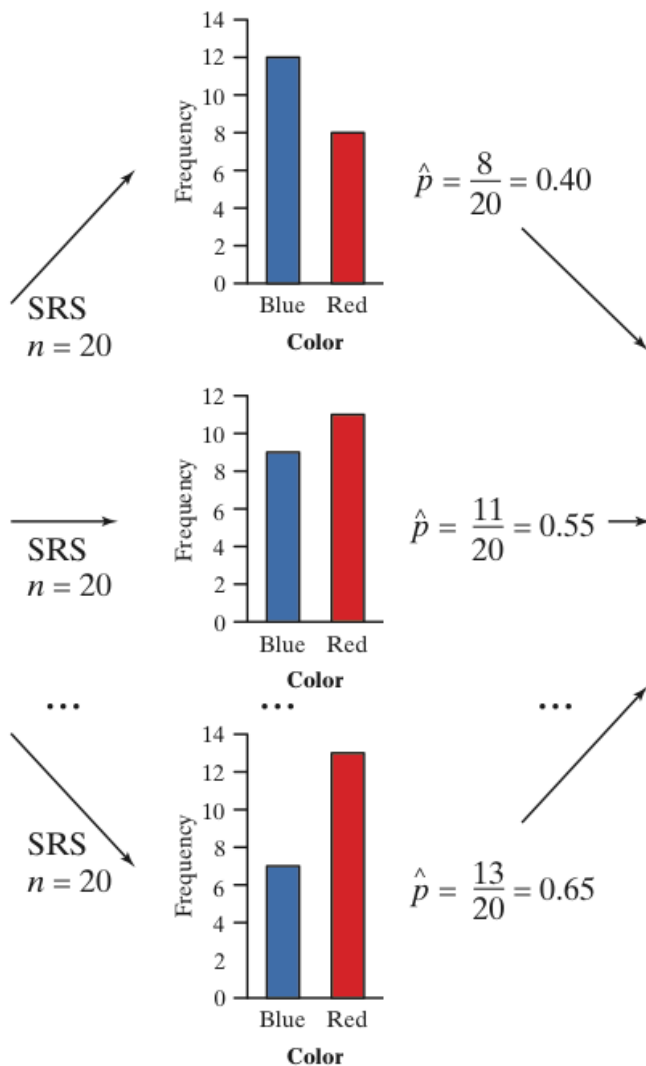
A sampling distribution describes how a statistic varies in many samples from the population.

Population distribution

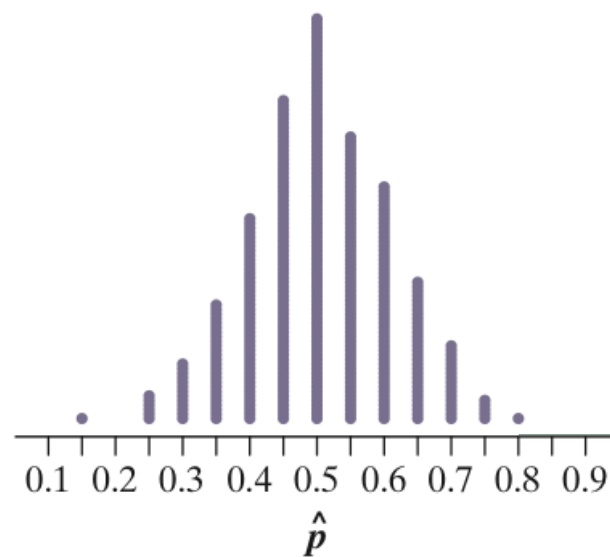


Parameter: $p = 0.50$

Distributions of sample data



Sampling distribution



Sampling distribution: Illustration

Example

SAT Scores at WFU,

Population Distribution of Student Scores:

$$\mu_x = 1300, \sigma_x = 80$$

Sampling Distribution of 25 WFU Student Scores

$$\mu_{\bar{x}} = \mu_x = 1300$$

$$\sigma_{\bar{x}} = \sigma_x / \sqrt{N} = 80/5 = 16$$

General Equation

In general,

- $\mu_{\bar{x}} = \mu_x$
- $\sigma_{\bar{x}} = \frac{\sigma_x}{\sqrt{N}}$
- $\sigma_{\bar{x}}^2 = \frac{\sigma_x^2}{N}$

Because the mean of the statistic $\bar{x}$ is always equal to the mean μ of the population

- (that is, the sampling distribution of $\bar{x}$ is centered at μ),
- we say the statistic $\bar{x}$ is an unbiased estimator of the parameter μ .
- Note: on any particular sample, $\bar{x}$ may fall above or below μ .

The sampling distribution of $\bar{x}$ (illustrated)



Population,
mean $\mu = 25$

$$\left. \begin{array}{l} \text{SRS size } 10 \rightarrow \bar{x} = 26.42 \\ \text{SRS size } 10 \rightarrow \bar{x} = 24.28 \\ \text{SRS size } 10 \rightarrow \bar{x} = 25.22 \\ \vdots \end{array} \right\} = 25$$

Sampling distribution,
mean $\mu_{\bar{x}} = 25$

Standard Deviation of the Sampling Distribution

Because the standard deviation of the sampling distribution of $\bar{x}$ is $\sigma/\sqrt{n}$,

- the averages are less variable than individual observations, and
- averages are less variable than the results of small samples.

Standard Deviation of the Sampling Distribution



Not only is the standard deviation of the distribution of $\bar{x}$ smaller than the standard deviation of individual observations,

- it gets smaller as we take larger samples.
- The results of large samples are less variable than the results of small samples.

Although the standard deviation of the distribution of $\bar{x}$ gets smaller, it does so at the rate of $\sqrt{n}$, not n .

- To cut the sampling distribution's standard deviation in half, for instance,
 - you must take a sample four times as large, not just twice as large.



Wrapping Up...



This Time...
Laws that
describe the
shape of
distributions

Laws that describe the shape of distributions



Law of Large Numbers



Central Limit Theorem



The Law of Large Numbers

Although a single statistic, like $\bar{x}$, is rarely identical to the population mean, we can still use it as a reasonable estimate of the population mean.

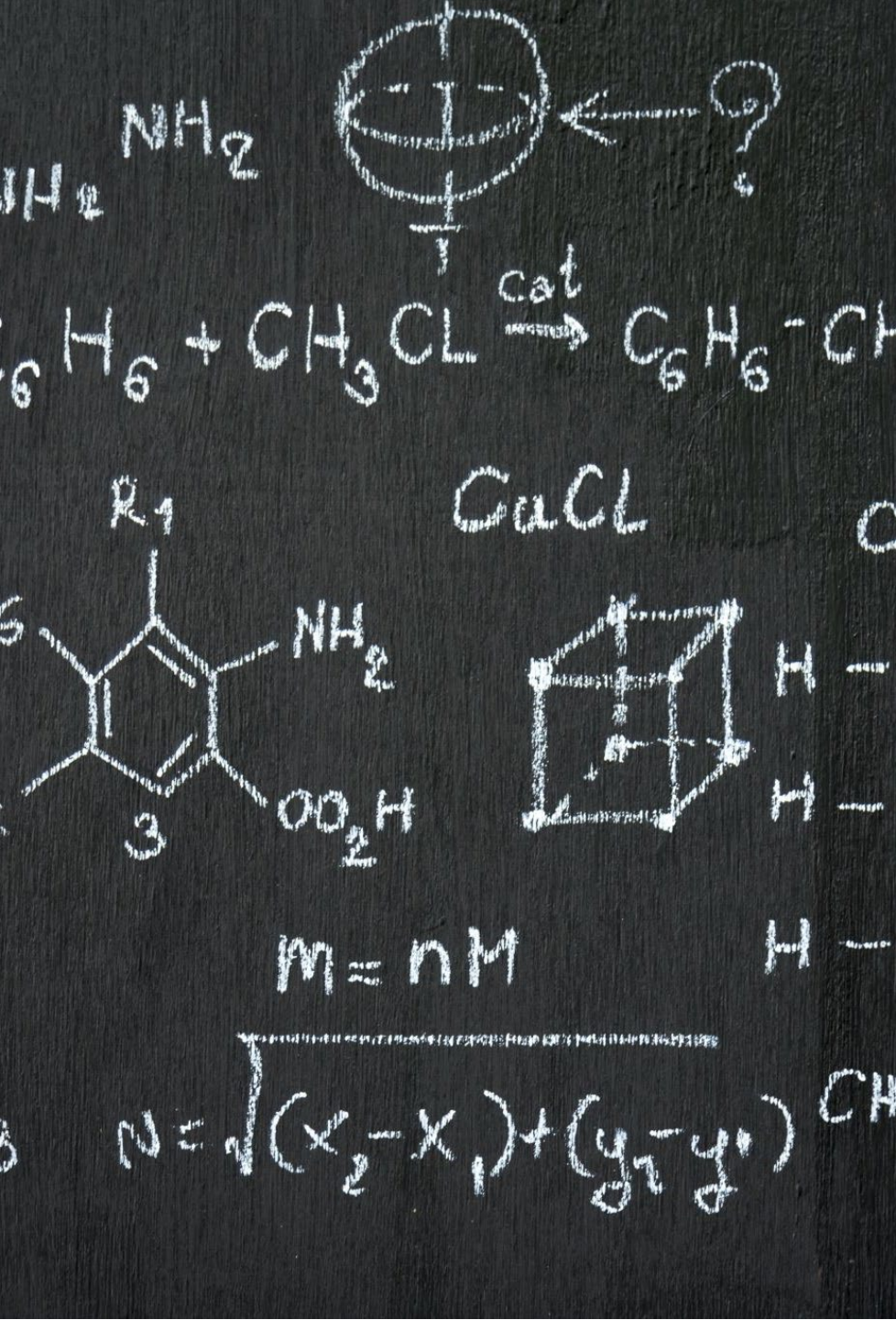
Why?

If we keep taking larger and larger samples, the statistic $\bar{x}$ is guaranteed to get closer and closer to the parameter μ .



The Law of Large Numbers

- Draw observations at random from any population with finite mean μ .
- As the number of observations drawn increases,
 - the mean $\bar{x}$ of the observed values tends to get closer and closer to the mean μ of the population.



Central Limit Theorem

Similarly, the Central Limit Theorem describes the shape of a sampling distribution.

Draw an SRS of size n from any population with mean μ and finite standard deviation σ .

- The central limit theorem says that when n is large, the sampling distribution of the sample mean $\bar{x}$ is approximately Normal:
- $\bar{x}$ is approximately $N\left(\mu, \sigma/\sqrt{n}\right)$

Central Limit Theorem

No matter what the shape of the raw score distribution.

The central limit theorem allows us to use normal probability calculations to answer questions about sample means from many observations,

- even when the population distribution is not normal (e.g, skewed, uniform, bimodal)

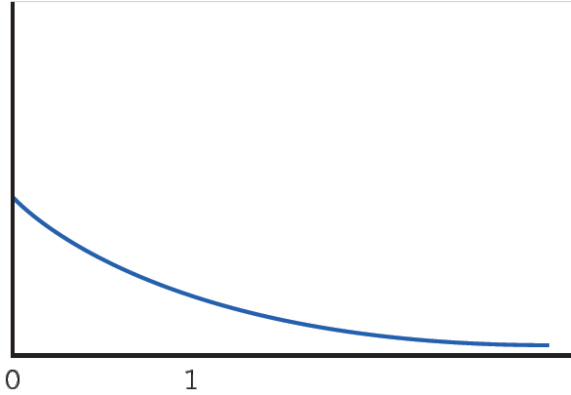
Reiterate

With a simple random sample,

A sampling distribution of the mean is normal if:

- The raw score distribution is normal;
OR
- The sample size is “large”
 - $N = 30$, is usually large enough
 - The central limit theorem does not apply under small sample size

Central limit theorem: example



Based on service records from the past year, the time (in hours) that a technician requires to complete preventative maintenance on an air conditioner follows the distribution that is strongly right-skewed and whose most likely outcomes are close to 0. The mean time is $\mu = 1$ hour and the standard deviation is $\sigma = 1$.

Your company will service an SRS of 70 air conditioners. You have budgeted 1.1 hours per unit. Will this be enough?

The central limit theorem states that the sampling distribution of the mean time spent working on the 70 units has:

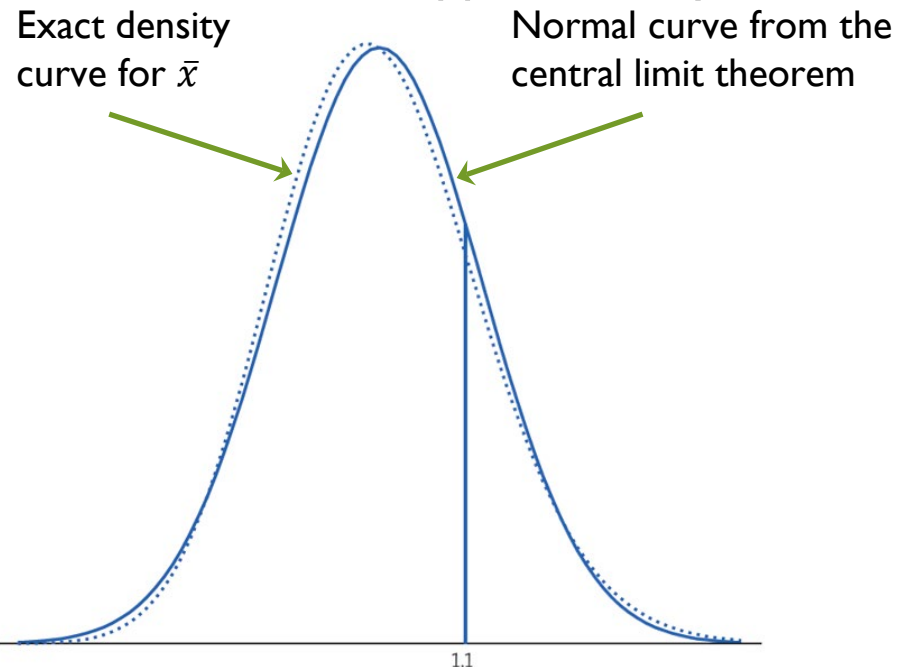
$$\mu_{\bar{X}} = \mu = 1,$$
$$\sigma_{\bar{X}} = \frac{\sigma_X}{\sqrt{n}} = \frac{1}{\sqrt{70}} = 0.12,$$

and a Normal distribution shape.

Central limit theorem: example

Your company will service an SRS of 70 air conditioners. You have budgeted 1.1 hours per unit. Will this be enough?

The sampling distribution of the mean time spent working is approximately $N(1, 0.12)$ since $n = 70 \geq 30$.



$$z = \frac{1.1 - 1}{0.12} = 0.83$$

$$\begin{aligned} P(\bar{x} > 1.1) &= P(Z > 0.83) \\ &= 1 - 0.7967 = 0.2033 \end{aligned}$$

If you budget 1.1 hours per unit, there is a 20% chance the technicians will not complete the work within the budgeted time.

Wrapping
Up...

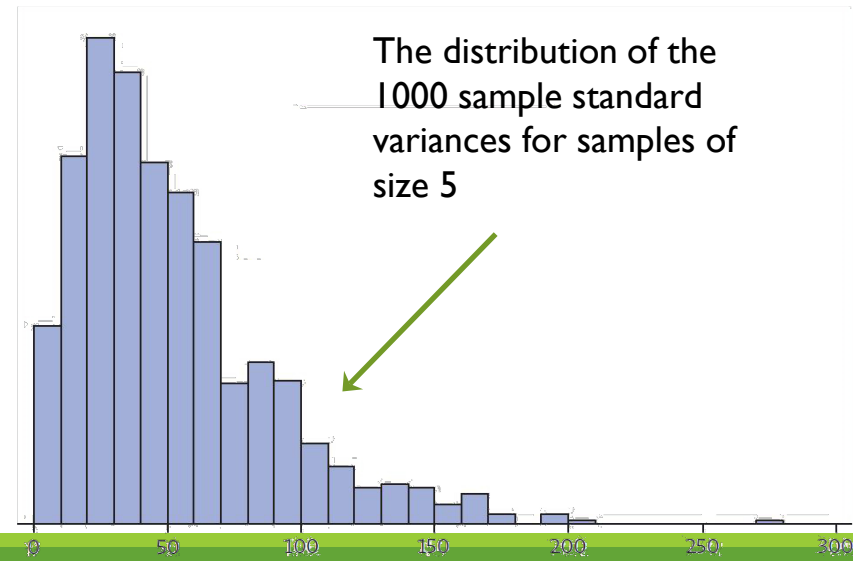
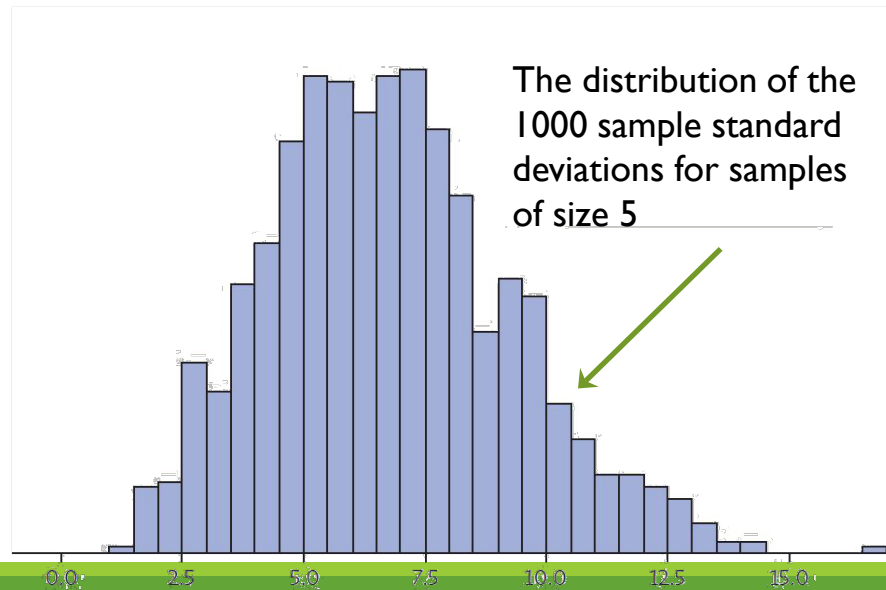


The background features a dark gray grid with a pattern of small, light brown dots. A thick, dark blue wavy line, resembling a signal or waveform, runs horizontally across the middle of the image. The title text is centered over this graphic.

Sampling distributions & statistical significance

Sampling distributions and statistical significance

- We have looked carefully at the sampling distribution of a sample mean.
- However, any statistic we can calculate from a sample will have a sampling distribution.





The sampling distribution of a sample statistic is determined by:

- the particular sample statistic we are interested in,
- the distribution of the population of individual values from which the sample statistic is computed, and
- the method by which samples are selected from the population.

Sampling distributions and statistical significance

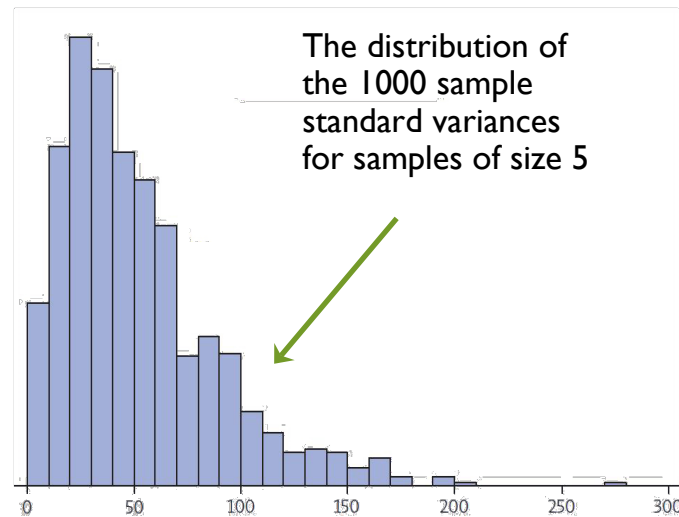
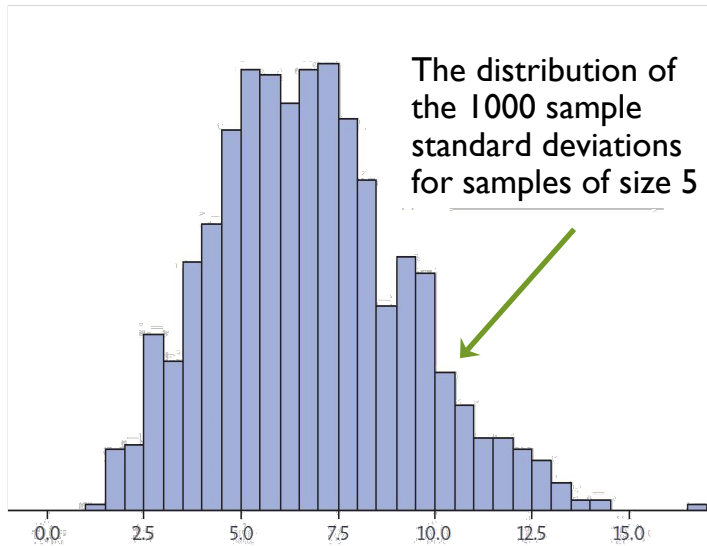


The sampling distribution allows us to determine the probability of observing any particular value of the sample statistic in another such sample from the population.



We said that an observed effect so large that it would rarely occur by chance is called **statistically significant**.

Consider This...



We may decide, based on our observed set of 1000 samples, that because we say only 2 with variances above 200, that that is a statistically significant event.

A photograph of a woman with a bindi on her forehead, smiling as she applies colorful paint to a young child's face. The child has their eyes closed and is wearing a pink floral shirt. The woman is wearing a blue patterned shirt. The background is a plain, light-colored wall. The text "Wrapping Up..." is overlaid in white, centered horizontally, with thin white lines above and below it. A solid green bar is at the bottom of the image.

Wrapping Up...